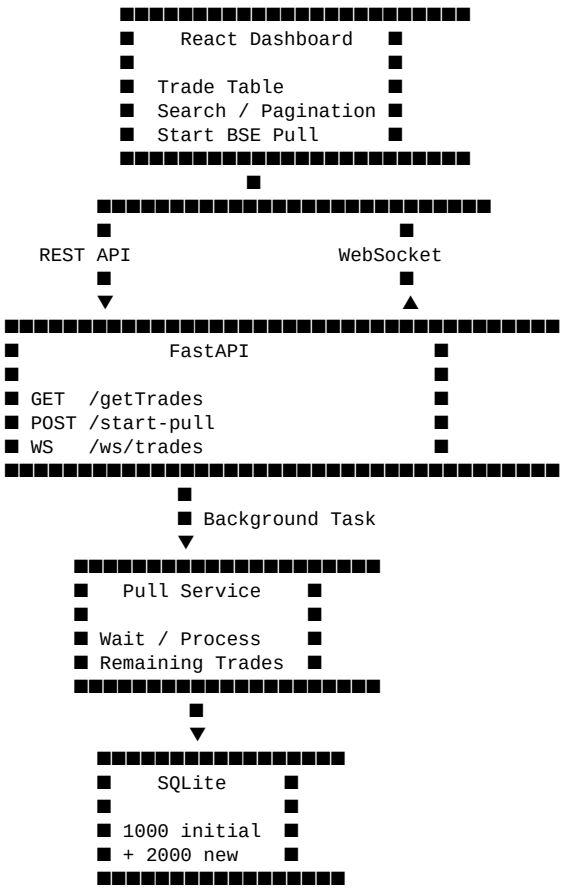
Architecture Note — Mock BSE Trade API & Real-Time Dashboard

Architecture Diagram



Why This Design?

1. Background Task

The simulated BSE API can take up to **15 minutes**, while the network may terminate HTTP connections that remain open for more than 30 seconds.

Therefore, `/start-pull` only starts the background task and **returns immediately**. The long-running operation continues independently.

2. SQLite

SQLite stores the trades that have already been pulled.

• Initially: **1,000 trades**
• After the pull: **3,000 trades**

This also allows the dashboard to display already available data immediately.

3. REST API

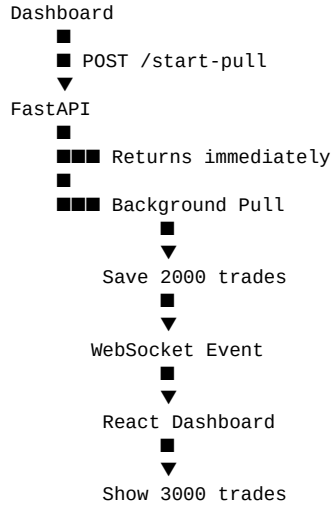
REST is used for normal request/response operations:
• `GET /getTrades` → retrieve available trades
• `POST /start-pull` → start the long-running pull

4. WebSocket

The dashboard maintains a WebSocket connection with the backend. When the background pull finishes, the backend sends a notification through WebSocket. The frontend then fetches the latest trades.

This avoids a **polling loop** and eliminates the need for a page refresh.

Overall Flow



This design keeps the dashboard responsive while safely handling a long-running trade-data operation.